Laiba Anwar

MS Bioinformatics Student | Biotechnology Graduate
Email: Laibaanwar549@gmail.com | Nationality: Pakistani

PROFILE

MS Bioinformatics student with a BS in Biotechnology, combining hands-on wet-lab experience (PCR, qPCR, ELISA, microbiology) with computational analysis skills. Experienced in molecular biology, microbiology and statistical data analysis, with an independent research project and internships at national research institutes.

EDUCATION AND TRAINING

MS Bioinformatics, National University of Science and Technology (NUST), Islamabad, Pakistan
07 Oct 2026 – Present

BS Biotechnology, COMSATS University Islamabad, Abbottabad Campus, Pakistan
01 Mar 2021 – 27 Jan 2025 | Final grade: 3.25/4.00

- Built a solid foundation in molecular biology, genetics, microbiology, biochemistry and cell biology.
- Studied advanced topics including recombinant DNA technology, cell and tissue culture, and pharmaceutical biotechnology.
- Developed practical laboratory skills through basic and analytical techniques in biotechnology.
- Strengthened research, analytical thinking and problem-solving skills through laboratory training and coursework.

FSc (Pre-Medical), Islamabad Model College for Girls (Post Graduate), G-10/4, Islamabad, Pakistan
30 Jul 2020 | Final grade: A

Matric (Science), Govt Girls High School, Dhoke Hasso, Rawalpindi, Pakistan
21 Jul 2018 | Final grade: A+

SKILLS

Molecular biology: DNA and RNA techniques (PCR, qPCR), DNA extraction, gel electrophoresis, ELISA (Enzyme-Linked Immunosorbent Assay)

Microbiology: bacteriology, media preparation, culturing of microbes

Other laboratory skills: tissue culture, laboratory animal handling, vaccine quality testing

Data and tools: data interpretation, biostatistics, bioinformatics and databases, Microsoft Office, Canva

Soft skills: communication, teamwork, organisational and planning skills, leadership, management

WORK EXPERIENCE

Intern, National Institute of Genomics and Advanced Biotechnology (NIGAB), NARC, Islamabad, Pakistan
25 Feb 2025 – 08 Apr 2025

- Gained hands-on experience in molecular biology techniques, including DNA and RNA extraction, PCR and qPCR.
- Studied the principles of shotgun sequencing.
- Contributed to plant biotechnology projects, including fieldwork, plant transfer and sample preparation.
- Worked safely in a laminar flow hood and followed best practice for aseptic and analytical procedures.
- Developed practical skills in experimental planning, molecular analysis and laboratory documentation.

Intern, National Veterinary Laboratories (NVL), Islamabad, Pakistan
15 Aug 2023 – 14 Sep 2023

- Conducted molecular biology experiments, including PCR, ELISA and gel electrophoresis.
- Carried out microbiological and bacteriological procedures, including culture handling and media preparation.
- Assisted with animal handling, vaccine quality testing and antibiotic selection procedures.
- Developed laboratory skills in experimental setup, data recording and analytical techniques while following strict safety and ethical protocols.

Distributor Coordinator, Pakistan Microbiological Associates (PMA), Rawalpindi, Pakistan
09 Sep 2025 – 13 Dec 2025

- Managed communication with customers, suppliers and internal teams, strengthening communication and interpersonal skills.
- Assisted in preparing tenders, quotations and official correspondence, and supported procurement and distribution of laboratory instruments and diagnostic kits.
- Managed online documentation and record keeping, and provided administrative support, developing leadership, problem-solving and teamwork skills.

PROJECTS

Stimulatory Effect of Selected Amendments on Growth Promotion of Soybean (*Glycine max*)
2024

- Conducted an independent research project investigating the effect of selected soil amendments on soybean growth.
- Designed and executed experiments, including treatment setup, sample monitoring and data collection.
- Performed complete data analysis using statistical methods such as ANOVA, created graphs to visualise results, and compiled a detailed scientific report.
- Strengthened skills in experimental planning, laboratory techniques, statistical analysis, data interpretation and scientific documentation, carrying out a full research study independently from conception to reporting.

HONOURS AND AWARDS

CAMS (COMSATS Abbottabad Merit Scholarship), COMSATS University Islamabad, Abbottabad Campus
01 Mar 2021

- Awarded for outstanding academic performance, maintaining merit for the full four-year scholarship period.

CERTIFICATIONS

Certificate of Completion: Workshop on Bioinformatics for Microbiology
American Society for Microbiology, 21 Aug 2024 (online)

CONFERENCES AND SEMINARS

- Pakistan's Digital Health Ecosystem: Challenges and Opportunities, 10 Aug 2024
- Pathway to Collaborative Networking Series: Applications of OMICS Technologies in Health and Disease, 30 Jul 2024
- Microbial Diversity in a Changing Climate, 30 Mar 2021

NETWORKS AND MEMBERSHIPS

- Pakistan Biological Safety Association, Islamabad (May 2024 – Present)
- American Society for Microbiology, United States (May 2024 – Present)

VOLUNTEERING

Décor Team Volunteer, Biotech Society, COMSATS University Islamabad, Abbottabad Campus
2024

Volunteered on the decoration team during university sports week, assisting with venue decoration, coordination and creative design, and helping prepare scientific charts, models and displays showcasing departmental activities and concepts.

Volunteer, Biotech Society, COMSATS University Islamabad, Abbottabad Campus
2024

Received a Certificate of Excellence for outstanding commitment and dedication as a volunteer during society activities and events.